

SHOW ME YOUR AGENT / NUS-ISS

A little more order in your kitchen.

Adaptive restaurant inventory and procurement agent

ReStock.

Keep purchasing recommendations valid as reality changes. ReStock investigates new sales, inventory and supplier evidence, calculates a supported response, and leaves purchasing authority with the restaurant manager.

The core design commitment

AI chooses what needs investigation. Deterministic services calculate and validate quantities, feasibility and costs. Backend state and exact-version manager approval remain authoritative.

Submission identity	Details
Team identity	Team 2B22 / NW0H3J8Y
Project	ReStock
Team / ownership	Chun Yang: Backend; Aniq: ML and numerical decision engine; Rudy: Agent/AI; Ethan: frontend and manager experience.
Repository	github.com/rudybrrr/restock-ai
Baseline	Main commit eba56bf · 27 September 2026
Evidence scope	Integrated local + verified hosted normal path; synthetic scenarios; live organiser-gateway normal/sales evidence documented.

This technical document covers architecture, safety, calculations and dated verification. The organiser Lightsail deployment has verified authenticated manager access and a supported live normal assessment to REVISE_PLAN / ENGINE / PENDING_APPROVAL. Hosted exact-version approval, safe escalation, restart persistence and real restaurant impact remain unverified.

01 / GOAL & SCOPE DEFINITION

A recommendation is only useful while its assumptions hold.

Business problem, intended users and the implemented MVP boundary

A restaurant manager must buy enough ingredients for service without tying up unnecessary cash or letting perishable stock expire. The difficult part is not just producing a forecast: yesterday's plan can become wrong after a promotion, a supplier change, a short delivery or unexpectedly high sales. The manager must reconcile facts scattered across sales, stocktakes, purchases and supplier terms.

Our product thesis

ReStock maintains an evidence-backed purchasing recommendation, rather than a one-off shopping list. It asks five practical questions: **what to buy, how much, when it can arrive, which approved supplier to use, and whether changed conditions require a new decision.** [2,3]

Designed for a small restaurant manager

- **During service:** report complete sales intervals, inspect recipe-based usage and distinguish estimates from physical counts.
- **When buying:** review quantities, supplier terms, arrivals and cash; approve an exact version; arrange the purchase outside ReStock.
- **When stock arrives or conditions change:** record receipts, delays and shortfalls, then follow the resulting assessment.
- **At closing:** submit final sales and per-lot physical counts, retaining corrections and history.

What this submission demonstrates

A synthetic five-dish, eight-ingredient restaurant with three approved suppliers. The normal connected procurement slice covers 16 February 2026 service. Bounded post-purchase contingency preserves existing commitments and recommends only additional quantities. Normal first-slice forecasts and projections are available through persisted manager-safe reads. [4,6,7]

Business value is a hypothesis, not an invented result

The intended benefit is fewer avoidable shortages, less unnecessary buying and less manual reconciliation. No real SME pilot, realised savings, waste reduction or intervention reduction has been measured. The evidence demonstrates decision correctness and workflow safeguards, not commercial impact.

02 / MANAGER WORKFLOW

From observations to an approved purchasing decision.

One workspace, explicit stages and no hidden purchase automation

Area	What the manager can do
Today	See the next action, pending recommendations, recent assessments, closing status, incoming deliveries and expiring-stock summaries.
Daily operations	Inspect physical inventory and sales-based estimates; report/correct sales; inspect stored normal forecasts and stock projections; draft, submit or correct closing updates.
Purchasing	Review versioned recommendations and evidence; approve/reject the exact version; record actual purchases; record partial/full receipts and supplier delivery changes.
Activity	Request an assessment, follow queue/processing/completion, inspect outcome, routing, evidence gaps and event/audit history.
Settings	Inspect menu/recipe and ordering references; compare supplier terms; record supplier changes, promotions and ordering decisions.

The four purchasing stages

Review → approve → record actual purchase → receive. These are separate operations. A recommendation is not an external commitment. Approval does not place an order, create a delivery or add inventory. Only a recorded receipt establishes arrived stock. [3,4]

Numbers keep their meanings

- **Physical count:** a staff observation at a specified time, retained per inventory lot.
- **Estimated inventory:** a derived balance through a stated cutoff; it is not a new count.
- **Expected supply:** only the unreceived outstanding commitment; received stock is not added twice.
- **Proposed supply:** hypothetical additions from an exact candidate/version, never presented as purchased or received stock.

First-login onboarding offers a skippable spotlight tour, with replay and per-area task guidance. Subtabs separate related work rather than placing every form and evidence panel on one long page. Unknown, loading, stale, failed and empty results remain distinct. [8]

03 / ARCHITECTURE & REASONING LOOP

Separate reasoning, calculation and operational authority.

A custom bounded Agent system, not an LLM-controlled database

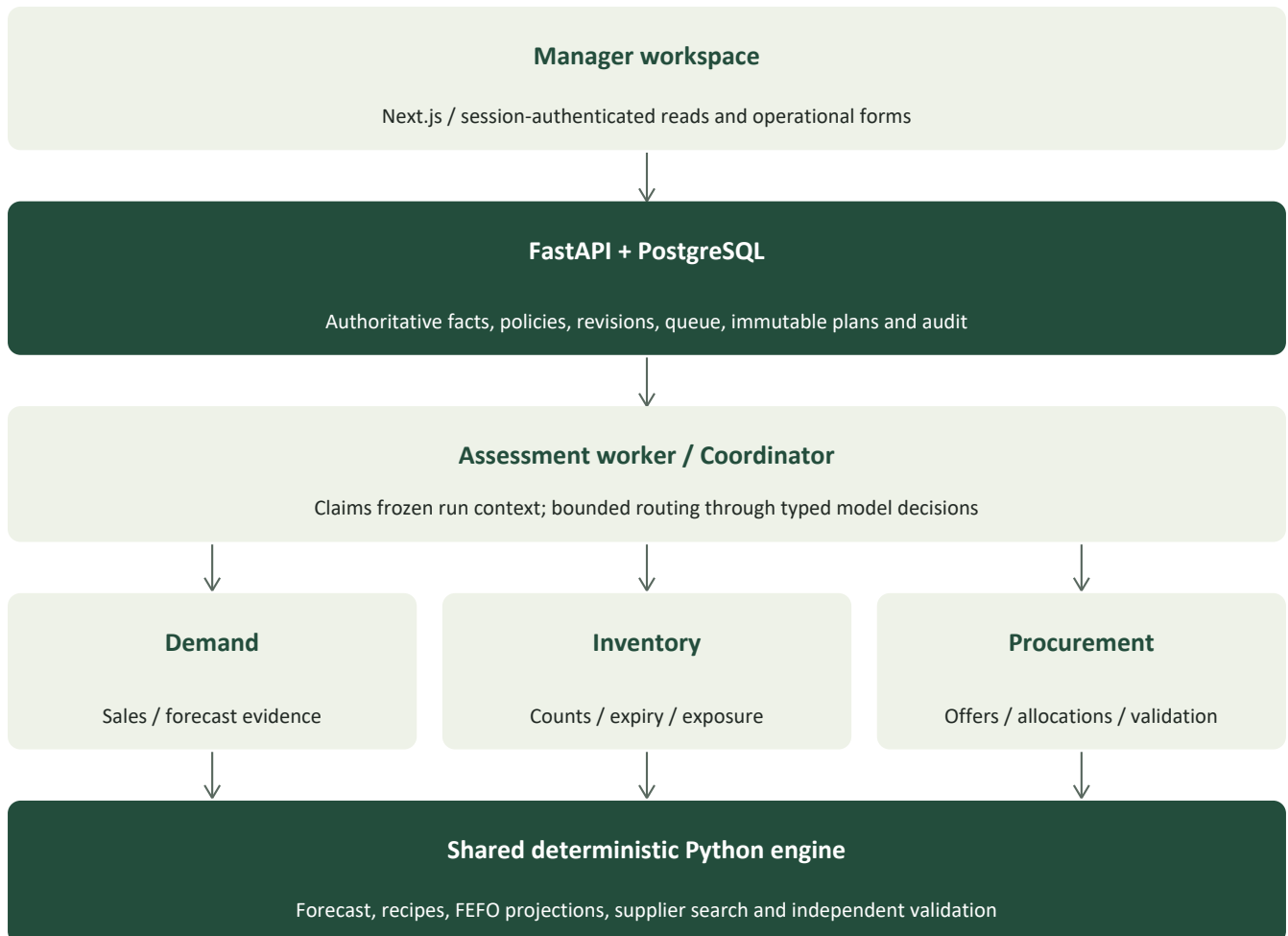


Figure 1. Authoritative data flow. The frontend never receives an Agent token. Specialist outputs return through Backend validation before a new plan is published. External purchase placement remains outside this system. [3]

Why three specialists?

Demand, Inventory and Procurement have different evidence needs and exact tool permissions. The Coordinator chooses the necessary investigations and bounded follow-ups; specialists cannot invoke one another. All use shared kernels rather than duplicating forecasting or optimisation logic.

State and memory

PostgreSQL retains operational facts, revisions, immutable policy/domain versions, captured run inputs, plan history and audit evidence. Each run binds **as_of** (operational cutoff), **known_at** (knowledge cutoff) and the captured state revision. The Agent cannot replace missing authoritative policy with prompt defaults. [3,4]

04 / AGENT DECISIONS & TOOL INTEGRATION

Investigate only what the decision needs.

Bounded routing over typed interfaces and deterministic tools

Component	Purpose-fit investigations	Authority boundary
Coordinator	Select specialist(s), validate identities and evidence references, synthesise bounded completion.	No direct approval or database mutation.
Demand	Sales context, forecast comparison, promotion evidence, uncertainty and reporting coverage.	No invented sales or materiality arithmetic.
Inventory	Counts, requirements, FEFO balances, expiry, incoming supply and stock-risk evidence.	No replacement of physical observations.
Procurement	Supplier options, allocation enumeration, optimisation, independent validation and approval requirement.	No arbitrary supplier, price, order or purchase writes.

Normal assessment path

A manager request or qualifying event queues a run. The worker claims it and freezes authoritative inputs. Typed specialist decisions select allowlisted tools. Procurement searches the approved finite domain, independently validates a candidate, and returns its resolvable reference. Backend rechecks revision freshness and publishes a feasible new version as **PENDING_APPROVAL**. [3,5,12]

Bounded execution

The Coordinator permits at most **two specialist rounds** and **six specialist calls**, with at most one external retry. Procurement has a six-logical-tool-call bound and one tool retry. Gateway decisions must be a single schema-valid JSON object with exact run/task binding; malformed/schema output receives at most one repair. Ambiguous output is not cherry-picked into a decision. [12]

An outcome is not an execution status

KEEP_CURRENT_PLAN retains a supported plan. REVISE_PLAN creates a validated revision for approval. ESCALATE records why automation cannot safely proceed. A run can be SUCCEEDED while its business outcome is ESCALATE: execution completed, but no purchase was authorised.

The live application worker uses the organiser gateway for typed specialist decisions. OpenClaw is an isolated smoke-test path, not the production application worker. Provider-free scripted routing is used separately for reproducible evaluation. [5,9]

Exact inputs. Explicit policies. Independently checked outputs.

Forecasts, recipe conversion, inventory exposure and cash-slice purchasing

Demand becomes ingredient demand

The seasonal baseline selects eligible historical daily revisions visible at issue time, excluding promotional/censored observations. It uses matching weekdays or an eligible-history fallback; insufficient history is unknown and requires manual input, not zero demand. Recipe quantities convert dish expectations into ingredient requirements. Service profiles allocate demand into dated half-hour intervals. [10]

Inventory exposure respects time and provenance

FEFO allocation uses expiry, receipt time, then lot ID. Only arrived, unexpired stock is usable; stock remains usable through its expiry date in Singapore time. Projections distinguish required from actually allocated usage, unmet demand, expiry and closing balance. A late arrival cannot erase an earlier shortage. Proposed purchases remain separate from fixed commitments. [6,7,10]

Frozen normal policy	Implemented first-slice meaning
Objective / horizon	CASH_SLICE_V1; 16 February 2026 service only, ending 21:00 SGT. No next-cycle coverage guarantee.
Issue / service	Issue: 15 February, 22:00 SGT. Lunch 11:00-14:00 (40%); dinner 17:00-21:00 (60%).
Budget / buffers	S\$100 new-order budget; zero safety stock for all eight ingredients; explicit ingredient-specific storage limits.
Supplier domain	Fresh, Pantry and Market; 24 frozen offer revisions and 24 dated opportunities, not supplier names alone.
Fees / risk	Once per supplier + arrival group; normal-only orders; historical reliability is context only.

Search checks MOQ, pack sizes, capacity, lead/cutoff timing, arrivals, expiry, storage, safety and budget under the frozen contract. Candidate feasibility and search completeness are separate: a diagnostic incumbent from incomplete search is not an actionable recommendation. [4,6,12]

Scope discipline: numerical waste-accounting and bounded 21-day economic modules are implemented, but the connected MVP is not a full waste/stockout/lost-sales economic optimiser. No browser fallback or Agent-generated policy activates those modules. [10]

06 / DEMONSTRATED END-TO-END BEHAVIOUR

Keep the plan. Add only the gap. Escalate when unsupported.

Synthetic scenarios with evidence levels stated separately

A / Normal recommendation: live organiser-gateway path

The 26 September validation records a fresh seeded run at the approved issue time. Ten valid typed gateway decisions led to deterministic supplier enumeration, optimisation and validation. Backend published **REVISE_PLAN / ENGINE / PENDING_APPROVAL**. The model selected investigations; it did not supply authoritative order quantities. [5]

B / Delivery shortfall: preserve existing commitments

A separate controlled baseline contains a 10 kg original purchase: 6 kg received in opening stock and 4 kg due later. A recorded rescue commitment initially supplies 4 kg for the 10 kg service need. The real deterministic post-purchase worker returns KEEP. When the rescue commitment falls to 2 kg, it recommends only the uncovered 2 kg of vegetables. [5,6]

Additional-only result: 2.000 kg vegetables, S\$11 new cash

Existing usable stock 6 kg + timely fixed rescue 2 kg = 8 kg against 10 kg need.

Additional requirement = 2 kg.

New purchase cash = 2 kg × S\$2/kg + S\$3 delivery + S\$4 emergency fee = **S\$11**.

The CONTINGENCY_ENGINE version remains PENDING_APPROVAL; existing purchases and receipts are not rewritten or duplicated.

The contingency baseline used controlled typed reasoning; the subsequent KEEP and additional-purchase assessments used the actual worker and deterministic contract. This is not a claim that every step was a live model call. Its S\$11 is new cash, not total economic cost or realised savings. [5,6]

C / Material sales: no invented late normal order

Forty synthetic sales intervals produced a complete persisted materiality result before live specialist reconsideration. Procurement could not reopen the approved issue window. The final result was **ESCALATE / CALCULATION_INCOMPLETE / UNSUPPORTED_ISSUE_OPENING**, with no new plan. Approval of the invalidated prior version returned **409 PLAN_VERSION_STALE**. [5]

The 27 September regression preserves that deterministic incomplete reason when allocation enumeration returns it. The local manager screen subsequently displayed the correct safe-escalation reason instead of generic TOOL_FAILURE. [13]

Trace these demonstrations to recorded runs

Normal: 65de2f64-5fd4-44a8-bc1c-c0479a32e004

Sales escalation: c5bf8405-5e5f-4659-8a84-58d75f7c7a2f

Additional purchase: ed59d0f6-ac86-4367-ac70-2faa7a92005a

These identify the isolated rehearsals in [5], not records guaranteed to exist in a newly seeded database.

Automate investigation. Keep purchasing authority human.

Risk-calibrated autonomy with enforceable boundaries

Risk	Implemented safeguard	Evidence to inspect
Prompt injection / tool misuse	Exact domain tool allowlists; typed decisions; no general database or approval mutation tool.	Permission/offline-agent tests; specialist schemas.
Invented or missing facts	Backend-frozen policy/domain, clocks and revision; unknown data fails closed rather than becoming zero.	Captured inputs, findings and escalation.
Stale calculation / approval	Revision checks before publication; approval targets exact plan/version; stale requests rejected.	PLAN_VERSION_STALE; historical versions remain readable.
Unsafe optimisation result	Complete-search status separate from independent candidate validation; diagnostic incumbents non-actionable.	Validation references and search-limit tests.
Duplicate buying / stock	Fixed purchases preserved; additions only; receipts and outstanding commitments reconciled.	Contingency evidence and purchase/receipt history.
Provider / parser failure	Typed identities; strict single-object parsing; bounded repair/retry; safe failure/escalation.	Gateway/specialist regressions and recorded failures.

Credential and information separation

Manager sessions use HttpOnly cookies; session tokens are stored as hashes. Browser mutations require an explicitly allowed Origin. Agent bearer credentials are a separate role and do not confer manager approval rights. Hosted settings require HTTPS/secure cookies; local HTTP configuration is not represented as deployment security. [4,11,12]

Manager evidence is a whitelist: outcomes, validation, routing/tool metadata, clocks, references and gaps. It omits private prompts, scratchpads, raw frozen state, evaluator-only truth and secrets. No hidden chain-of-thought is exposed as an explanation. [3,4]

Security boundary, not a certification: tests demonstrate specific guardrails, not immunity to all attacks. Hosted credential rotation, restricted network access, monitoring and acceptance tests remain deployment responsibilities. [11]

Evidence a judge can follow, not just a successful screen.

Traceability, reproducible tests and honest benchmark limitations

Activity links the trigger, captured revision, specialist/tool calls, validation, final outcome, plan transition and approval attempt. Calculation artifacts bind canonical content with SHA-256 and exact run/candidate identities. Historical results stay readable with freshness warnings rather than recalculating silently on GET. [3,7]

Verification	Result / evidence level
27 Sep final recording gate	1,341 PostgreSQL backend tests passed on main eba56bf; Ruff clean; Pyright 0 errors / 0 warnings; frontend lint and production build passed. Fresh full-suite submission gate. [18]
ML publication checkpoint	864 numerical tests passed on the main-based numerical slice; Ruff, Pyright and formatting passed. No connected waste/economic acceptance implied. [10]
Connected frontend	Real API / PostgreSQL / worker-to-browser normal calculation checks, including queued-result refresh without reload; controlled specialist reasoning, desktop/mobile. [7,8]
Live model evidence	Local normal plan publication and material-sales safe escalation are documented via organiser gateway. Hosted normal run reached REVISE_PLAN / ENGINE / PENDING_APPROVAL. Contingency starts from a controlled baseline. [5,18]

Historical comparator: seven development scenarios

Provider-free adapter	Outcome correctness	Routing accuracy	Failed runs
Static	0.714	0.143	0 / 7
Rule	0.714	0.714	0 / 7
Adaptive / scripted	0.714	0.571	0 / 7

The 19 September harness ran 21 executions (seven scenarios × three adapters), with repeatable boundary fingerprints across two runs. The scripted Adaptive path missed one replan and made one unnecessary replan; it did not outperform Rule on routing. This small development selection is not a live-model or held-out quality score. Separate permission tests cover prompt injection; it is not part of the seven-scenario aggregate. [9]

Not measured: live availability, token/cost/latency distributions, real food waste, stockouts, lost sales, savings and deployment reliability. Passing tests are evidence of specified cases, not business outcomes.

Make the decision legible.

Keep technical evidence available.

A projection is meaningful only when its supply basis is visible

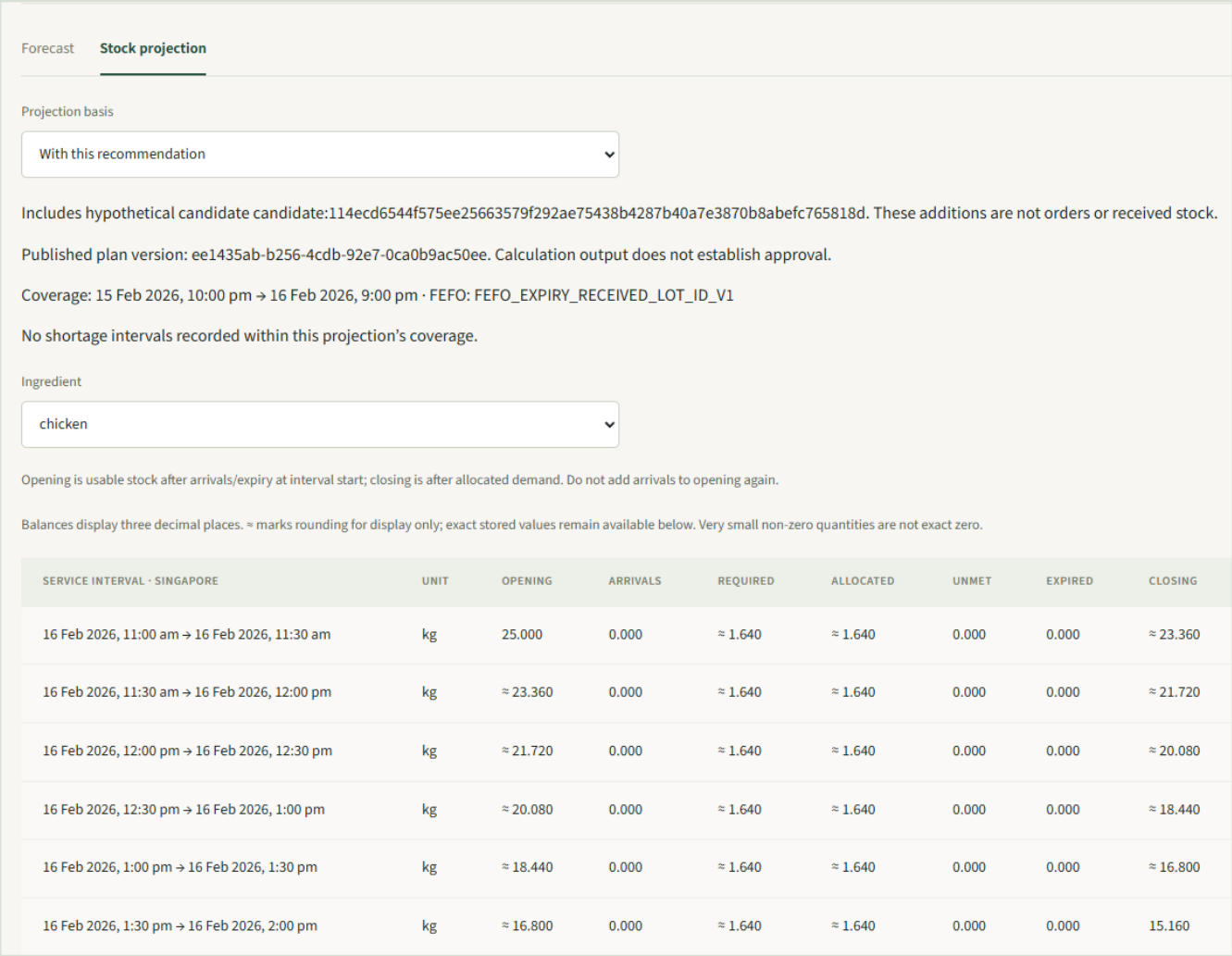


Figure 2. Connected normal-projection browser-test capture. The visible **Projection basis: With this recommendation** identifies hypothetical candidate supply; the text explicitly says these additions are not orders or received stock. Opening, required, allocated, unmet and closing quantities retain distinct meanings. [7]

This is a real persisted Backend artifact from an isolated synthetic database with controlled specialist reasoning, not hosted or general 21-day evidence. Display rounding does not replace exact stored values. No shortage in this candidate projection is not proof of sufficient physical stock without the proposed additions.

Five task areas, local subtabs and expandable evidence organise the workflow. The first-login spotlight tour can be skipped or replayed. Pending selections refresh automatically; loading, unknown, incomplete and empty states are distinct. Mobile checks preserve warnings and exact-version approval. [8]

A coherent prototype.

An explicit production boundary.

Runtime shape, operational readiness and next milestones

Implemented technology

Next.js/React manager UI; FastAPI/Pydantic typed HTTP contracts; SQLAlchemy, Psycopg and Alembic with PostgreSQL; a separate Python assessment worker; shared Decimal-based numerical kernels. The organiser gateway provides typed specialist decisions using the configured Claude Sonnet 4.5 route. Successful organiser-gateway calls are not independent proof of direct AWS/Bedrock hosting. [3,5,11,12]

Deployment requires four running components

Frontend + API + PostgreSQL + supervised worker. API and worker share the same database and authoritative configuration. Migrations run explicitly. The frontend is built for the public API origin. HTTPS, secure session cookies, exact allowed origins and server-side secrets are required. No Agent token or gateway key belongs in NEXT_PUBLIC variables. [11]

Mandatory organiser deployment: the latest supplied clarification requires deployment in the organiser's Lightsail environment with a live URL. It is not an optional alternative. The team provisioned **restock-demo** in the organiser lease, Singapore, on Ubuntu 24.04; Caddy terminates HTTPS. [16,17]

Live URL: <https://restock.13.250.80.71.sslip.io>
Verified: HTTPS, authenticated manager access and a hosted-worker supported live assessment to REVISE_PLAN / ENGINE / PENDING_APPROVAL. Not acceptance-tested on hosted: exact-version approval, safe escalation, and restart persistence/recovery. [17,18]

Remaining boundary	What is still required
Hosted acceptance	Verified: authenticated login/manager reads and supported live normal assessment to REVISE_PLAN / ENGINE / PENDING_APPROVAL. Still required for hosted acceptance: exact-version approval, safe escalation, restart persistence/recovery and credential rotation.
Observed waste	Approved Backend write/history/reversal lifecycle and chronological replay. Numerical helpers do not create a manager workflow; discrepancies must not be called waste.
Full economics	Freeze primary objective, terminal and continuation policies; persist certified request/results; connect safe Agent consumption and manager display. No actionable full-horizon claims yet.
Broader calculations	General multi-day forecasting and wider contingency display are beyond the connected normal first-slice screen. Unsupported clocks/domains remain incomplete.
Operational evidence	Real restaurant/POS integration and a pilot measuring waste, shortages, costs and staff effort; provider reliability and hosted recovery measurements.

A local PostgreSQL timeout previously required worker restart. Systemd supervision is configured on Lightsail, but recovery has not been acceptance-tested. The demo video and deployment URL/artifact are submitted separately. External POS sync and automatic supplier checkout remain outside scope.

11 / JUDGE NAVIGATION & REFERENCES

Review the evidence against the seven criteria.

Pinned source baseline and reproducible evidence map

Judging area [1]	Where to inspect in this write-up
Goal & scope	Pages 2-3: manager problem, intended value and bounded implemented scope.
Architecture & reasoning	Pages 4-5: authoritative state, specialist routing, typed decisions and bounded loop.
Tool use & integration	Pages 5-7: actual tools, frozen policies, deterministic calculations and validated publication.
Autonomy & human-in-the-loop	Pages 3,7-8: exact-version review, preserved commitments, additional-only purchases and escalation.
Safety, security & guardrails	Page 8: permissions, parser/revision checks, fail-closed data and credential separation.
Observability & evaluation	Pages 9-10: dated acceptance, honest comparator results, provenance and manager evidence.
Platform & tooling	Pages 4,11,13: custom worker/runtime, hosting limits and visible deployment evidence.

Evidence index

- [1] **ShowMeYourAgent Hackathon Briefing, release 1** - Supplied organiser PDF, pp. 13,16,18,21.
- [2] **Original ReStock v5 project concept** - Supplied project plan; intended direction, not implementation proof.
- [3] **docs/ARCHITECTURE.md** - Authority, Coordinator, specialists and lifecycle.
- [4] **docs/BACKEND_SPEC.md** - Operational facts, access, immutable history and policy contracts.
- [5] **docs/FINAL_VALIDATION_2026-09-26.md** - Named live normal/sales and deterministic contingency runs.
- [6] **docs/POST_PURCHASE_CONTINGENCY_CONTRACT.md** - Exact fixed-supply fixture, cash calculation and limitations.
- [7] **docs/MANAGER_CALCULATION_RESULTS.md** - Persisted immutable forecast/projection reads.
- [8] **docs/FRONTEND_CALCULATION_RESULTS_HANDOFF.md** - Connected refresh/browser evidence; UI tests under apps/web/tests.
- [9] **docs/evaluation/LOCAL_RESULTS_2026-09-19.md** - Provider-free comparator metrics and limitations.
- [10] **docs/ML_FRONTEND_HANDOFF.md** - Numerical scope, 864-test checkpoint and unresolved policies.
- [11] **docs/DEPLOYMENT_HANDOFF.md** - Runtime/configuration and hosted acceptance procedure.
- [13] Pull request #63 / main eba56bf - Reason mapping + UI alignment; focused regressions and pre-merge frontend checks.
- [14] ReStock ML / Decision Engine Plan v2 - Review-reconciled numerical design; unresolved policies retained.
- [15] ReStock ML v3 Integration Amendment - Catalogue alignment, source precedence and first-slice cash scope.
- [16] Latest organiser submission instructions - Separate Business Proposal / Technical PDFs, demo video and deployment evidence; organiser Lightsail/live URL required.
- [17] 27 Sep Lightsail operator screenshots - Four active services, local health, frontend build and HTTPS /login 200; public status capture reproduced on page 13.
- [18] Final demo recording / QA, 27-28 Sep - main eba56bf fresh 1,341-test backend gate, frontend lint/build, authenticated hosted manager session and supported live normal assessment.

Baseline: main eba56bf, 27 Sep 2026. Final engineering gate is fresh on that commit. Plans [2,14,15] document intent, not blanket policy approval. Hosted exact-version approval, safe escalation, restart persistence and real restaurant impact remain outside verified hosted acceptance.

APPENDIX / DEPLOYMENT EVIDENCE

A live endpoint. A clearly bounded claim.

Operator-provided organiser Lightsail evidence, 27 September 2026 [17]

Deployment URL: <https://restock.13.250.80.71.sslip.io/login>

Instance: restock-demo / Singapore / organiser-provided lease. This capture was supplied during deployment, not generated from a simulated browser response.

```
ubuntu@ip-172-26-13-249:~/restock-ai/apps/web$ sudo caddy fmt --overwrite /etc/caddy/Caddyfile
ubuntu@ip-172-26-13-249:~/restock-ai/apps/web$ sudo systemctl enable --now caddy
ubuntu@ip-172-26-13-249:~/restock-ai/apps/web$ sudo systemctl reload caddy
ubuntu@ip-172-26-13-249:~/restock-ai/apps/web$ sudo systemctl is-active restock-api restock-worker restock-web caddy
active
active
active
active
ubuntu@ip-172-26-13-249:~/restock-ai/apps/web$ curl --max-time 30 -I https://restock.13.250.80.71.sslip.io/login
HTTP/2 200
alt-svc: h3=":443"; ma=2592000
cache-control: s-maxage=31536000
content-type: text/html; charset=utf-8
date: Sun, 27 Sep 2026 13:33:11 GMT
etag: "wg46fo6gut6q6"
server: Caddy
vary: rsc, next-router-state-tree, next-router-prefetch, next-router-segment-prefetch,
accept-Encoding
x-nextjs-cache: HIT
x-nextjs-prerender: 1
x-nextjs-prerender: 1
x-nextjs-stale-time: 300
x-powered-by: Next.js
content-length: 8720

ubuntu@ip-172-26-13-249:~/restock-ai/apps/web$
```

Figure 3. The four **active** responses correspond to restock-api, restock-worker, restock-web and Caddy. The HTTPS request to /login returns **HTTP/2 200**; Caddy and Next.js headers identify the serving path. This demonstrates process liveness and public TLS reachability at the captured time.

Hosted acceptance verified

- Authenticated HTTPS manager login and reads using server-side credentials were verified without reproducing secrets.

- A supported hosted live organiser-gateway assessment was queued, completed by the hosted worker, and published REVISE_PLAN / ENGINE / PENDING_APPROVAL.

Still outside hosted acceptance

Hosted exact-version approval, hosted safe escalation, and persistence/recovery after restart were not acceptance-tested. Local evidence separately demonstrates safe escalation and stale-version rejection. The deployment remains a prototype, not a production-readiness or availability claim.